

Quality Study of CNT based Inkjet Printing

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Abstract— Despite the high potential of carbon nanotube (CNT) as a replacement material for many state-of-the-art devices, in most of the cases, the choice of incorrect composition for CNT ink leads to improper printing or print failure. In this paper, the authors have investigated the quality of inkjet printing of multi-walled carbon nanotube (MWCNT) films with different types of MWCNT inks to measure their sheet resistances and predict the gauge factor for a sheet strain gauge. Results are validated by comparing experimental values with the reference values found from an extensive literature review. Ink solution used for printing (having best sheet resistance) is selected and the resistance is measured for different number of layers for print.

Keywords— Inkjet Printing; MWCNT; SWCNT; Solvent; Sheet Resistance

I. INTRODUCTION

Carbon nanotube (CNT) are allotropes of carbon with a cylindrical nanostructure. CNT discovered in 1991[1]. Since its inception, it has gained popularity in industrial as well as research field due to its excellent mechanical strength, stiffness, electrical conductivity and thermal conductivity [2]. Due to high electrical conductivity, we use CNT extensively in electronic devices like transistors, radio-frequency identification tags, sensors, photonics and biological sensors [3]. CNTs are broadly classified into single-walled carbon nanotube (SWCNT) and multi-walled carbon nanotube (MWCNT) [4]. Conventional methods of CNT deposition on a substrate are dip coating, spray coating, electrophoretic deposition, screen printing, aerosol printing, transfer printing and contact printing [5,6]. Issues that have encountered with these methods are lack of homogeneity, thickness control of films and productivity in industrial application. To get rid of these issues researchers

attempted inkjet printing (IJP) as an alternate method to print CNT. IJP is a low-cost, quick printing method which requires no pre-fabrication of templates [6]. Additionally, it does not require any post-printing steps. Multiple ink cartridges can be used in parallel to print multiple materials with desired film thickness [6]. According to Magdassi, IJP can be broadly classified into drop-on-demand (DOD) and continuous printing methods. Between these two, DOD has gained popularity over the years. Despite offering advantages with CNT deposition, there are some issues reported with IJP. Major issues include bubble formation due to vaporization of ink [5], substrate selection which has a major role in determining the print quality and ‘coffee-ring’ effect [5]. The substrate must allow ink droplets to penetrate through the layers to ensure the longevity of the deposited films. Denneulin et al. used an SWCNT-COOH/PEDOT-PSS composite ink [6] to get rid of ‘coffee-ring’ effect. Both these factors encouraged researchers to experiment with various ink composition and find the best one to address the major shortcomings of IJP. MWCNT inks can be categorized as water-based ink, organic solvent-based ink, and functionalized ink. Ink solution must possess low viscosity and low surface tension to ensure good flowability and no clogging of the nozzle head. The solvent choice must be carefully done to ensure zero flocculation in a dispersed medium.

CNT molecules are hydrophobic. Due to high van der-Waals force prevailing between the CNT molecules, there is always a high possibility of flocculation in the dispersed medium. We added surfactants in water-based ink solution to counter these issues. Surfactants are amphiphilic molecules having a hydrophilic head and hydrophobic tail. When we introduce a surfactant in water-based CNT ink, molecules adsorb to the surface due to its hydrophobic tail. This creates a significant barrier around CNTs and prevents them from flocculation [6]. Sometimes these

surfactants also act as wetting agent [6]. It is important to control surfactant amount since the excess of this can cause a significant decrease in electric conductivity by creating thick barriers around CNTs and preventing minimum contact between layers required to conduct electricity. Water-based inks are easy to store, safe to handle.

Most of the organic solvents show low surface tension and low viscosity. Additionally, CNT molecules can easily dissolve in an organic solvent as it can bond with solvent molecules. There is no additional need of surfactant for organic solvent-based ink. According to researchers most organic solvent-based ink, the solution has a carbon nanotube concentration limit of approximately 0.1 mg/mL [6]. Though there are some drawbacks which include health hazards and volatility of the solvent. If solvent evaporates, leftover CNT particles may clog the nozzle. Though we can resolve both the issues by adhering to safety measures. The materials used for cartridges should not corrode the organic solvent used. This also limits the substrate selection for organic solvent-based CNT inks. Despite the difficulties, many research groups have successfully developed organic solvent-based ink using DMF and NMP [6]. In functionalized ink, we added a hydroxyl group(-COOH) to the sidewalls of CNT molecules through a series of chemical processes. Hydrophilic nature of carboxylic groups helps CNTs to dissolve in water. Use of sonication process is good for solvent selection. In this process, a controlled high-frequency wave for a limited time is sent through the solution to mix CNT molecules with the solvent. Following sonication, the ink is spun mechanically with high velocity. This collects the less heavy particles at the center as the centrifugal force causes heavier CNT particles to migrate away from the spinning axis. We can centrifuge the collected supernatant further if required. This is a way to avoid the presence of a heavier undissolved particle in the ink which can clog the nozzle head.

From our extensive literature study, we have found that there is a good scope of research in fabrication of inexpensive thermomechanical sensors using carbon nanotubes. And the most important factor that characterize a strain sensor is its sensitivity to deformation or the gauge factor. to know the sensitivity we need to know the surface resistance of CNT to electron flow. In this paper we have discussed the effect of surface resistance on different layers of CNT.

II. SHEET RESISTANCE AND GAUGE FACTOR

CNTs can be efficient enough in replacing copper interconnects beyond the 45nm technology nodes. If a CNT interconnects with reduced interconnect sheet resistance can be prepared, the implementation of CNT interconnects into CMOS process becomes feasible. We can consider a CNT AR scaling to 0.5 with all possible improvements producing a minimum sheet resistance of 50 Ω /sq as a viable replacement of copper interconnects [13].

From the literature review and online resources[15-18], we have found that the four-point probe method is extensively used to measure sheet resistance. This method has gained popularity over other proposed methods as it can measure very low resistance with precision. Especially, in the electronics industry, this method is very useful since they use lots of low resistance components.

Four-point probe set up consists of four equally spaced tungsten metal tip with finite radius [15]. The other end of the probes is supported by spring to minimize damage to the CNT layer during probing [15]. Illustration of the experimental set up is in Fig.1 and the experimental circuit is in Fig.2.

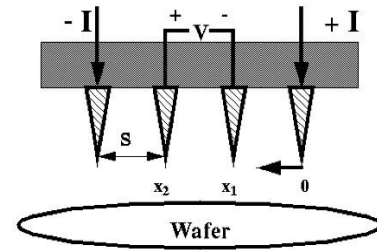


Figure 1. Schematic of Four-point probe set up [15]

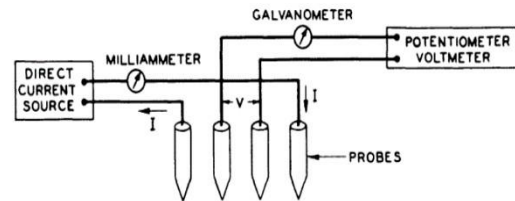


Figure 2. Schematic of the experimental circuit [16]

We can evaluate Sheet resistance of CNT layers with a thickness of less than 40% of spacing(S) [18] using

$$Rs = 4.53\left(\frac{\Delta V}{I}\right) \quad (1)$$

where I is the current applied between outer two probes, ΔV is a voltage drop between the inner two probes and Rs gives sheet resistance.

Complex and heterogeneous microstructures of composites have made it really difficult to monitor deformation in them by nondestructive testing techniques or sensors (like strain gauges). It is a good option to use Macroscopic aggregates of CNTs also called 'Bucky paper' or 'Buckytubes' for strain sensing due to its good piezoelectric property. As shown in previous work [14] CNT membrane based self-sensing GFRP (SGFRP) was used for tension-tension fatigue loading for an 8"x1" with a strain range of 0.002 - 0.009 and the piezoelectric response was recorded and the gauge factor was calculated using the relation

$$GF = ((R - R_0)/R_0)(1/\epsilon) \quad (2)$$

Where R is the electrical resistance at strained state and R_0 is the original resistance at un-strained state, ϵ is the longitudinal strain and gauge factor can be briefly defined as the relative change in electrical resistance to mechanical strain. The gauge factor (GF) is defined as the change in electrical resistance with respect to strain.

The gauge factors were found to be 9.34 at ascending and 5.08 at descending for the first few cycles and then the average sensitivity was 5.62 and 4.17 and delamination was observed at 80000 cycles. These results conclude that CNT based composite strain sensors have high sensitivity and fatigue life, and delamination due to fatigue being a primary failure mode in die packages makes these sensors highly viable. From the previous literature, we have found that there is a lack of obtaining gauge factors from different types of CNT inks and this gives a scope of research.

$$GF = \frac{\Delta R/R}{\Delta L/L} = \frac{\Delta R/R}{\epsilon} = 1 + 2\nu + \frac{\Delta \rho/\rho}{\epsilon}$$

Where ϵ is the strain which is $\Delta L/L$, ν is the Poisson's ratio and ρ is the resistivity. ΔR is the change in strain gauge resistance due to axial strain and lateral strain and R is the unstrained resistance of the strain gauge.

III. INK PREPARATION

The most important properties of ink for inkjet printing are good dispersion, low surface tension, and low viscosity.

The carbon nanotubes cling to each other by van der Waals force of attraction and in a liquid medium agglomeration and sedimentation becomes a very common phenomenon and this is highly undesirable because this may cause clogging in the inkjet nozzle. Therefore, the dispersion of CNT becomes necessary. This can be done by sidewall functionalization, organic solvents, and dispersants in the case of water-based ink. Dispersants not only prevent the sedimentation of CNTs but also give structural integrity to the nanotubes, they give highly concentrated dispersed CNT.

A. Water Based Ink

The dispersant/surfactant used here is dimethylformamide (DMF) [6].

The desired amount of MWCNT (1g) is added to a 100ml glass container with the addition of DMF next followed by the addition of deionized water. The container was then sonicated for 90 minutes and ideally to be centrifuged for 5mins at 12000rpm [6]. Then we collected the supernatant and measured the contact angle to confirm the surface tension was low enough. Contact angle measurement process is done as follows. The 2 μ L drop was placed on an untreated bar silicon substrate and the contact angle was measured using (FTA 125, First 10 Angstroms) it needs to be less than 15° for uniform and consistent printing [6]. The ink was then injected in the cartridge (HP 63 XL) and loaded in the printer (HP-desk jet 1112).

We took the following optimization. Every surfactant has a critical micelle concentration (CMC) value to maintain. If the concentration equals or exceeds the CMC value, the surface tension will be low and in the concentration is lower than the CMC value the surface tension will be higher which is not very desirable [6]. The CMC value of DMF in water is 2.37mg/ml, the ink should not be prepared below the concentration value. If excessive SDS is added, it prevents or decreases the number of CNTs dispersed per unit volume.

TABLE I. Optimized Value for Water Based Ink

MWCNT	10 mg/ml
SDS	7 mg/ml
SWCNT	0.8 mg/ml
SDS	3 mg/ml
CMC value	2.37 mg/ml

TABLE II. Results for Water Based Ink

SWCNT	SDS:CNT	5:1 to 10:1
MWCNT	SDS: CNT	0.3:1 to 1.5:1

B. Methanol Based Ink

The dispersant/surfactant used here is Methanol.

The multi-walled carbon nanotube brought from 1-gram CNTs was dispersed in 50 ml Methanol. We sonicated the mixture for 30 minutes in room temperature and ideally to be followed by centrifuge for 5 minutes to remove some impurities or possible remaining CNT aggregates [13]. The supernatant was collected. A small amount of the initial CNTs content was lost during this step. The final suspension obtained was opaque, black and stable.

IV. RESULTS AND DISCUSSION

The below table explains the nature of ink preparation steps for the different inks to be used in the experiment.

Solvent	Sonication	Centrifuge
Water	No (unfavorable dispersion characteristics)	
Methanol	Yes	No
Methanol	Yes	Yes
DMF & water	Yes	No
DMF & water	Yes	Yes

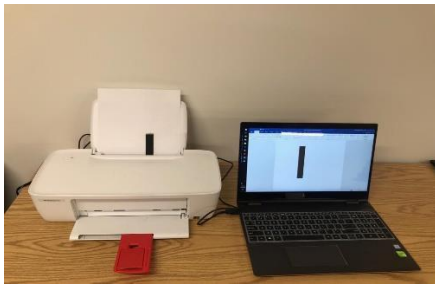


Figure 2. Experimental set-up showing Laptop connected to inkjet printer having CNT ink cartridges.

The experimental setup consists of a computer attached to an inkjet printer with an USB cable. The pattern or the circuit to be printed is created in a word document for printing. The compatible cartridges in the printer was cleaned and then filled with the prepared CNT ink. Multiple runs were done on the same paper with the same pattern to obtain the final output.



Figure 3. HP-63 cartridges used in the printer modified to contain prepared MWCNT ink.

The printed strips are then allowed to dry and observed for any coffee stain effect. The samples from methanol-based ink and DMF+water-based ink is as shown in figures 4 and 5 respectively.

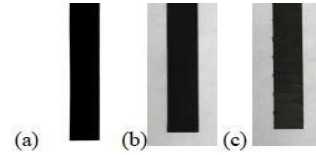


Figure 4. Printed CNT strips on paper from Inkjet printer with methanol based ink. (a) 5 runs (b) 10 runs (c) 15 runs.

The methanol based CNT ink gave the best electrical properties as expected from initial results. The values are shown in the separate plots for 5 layers and 10 layers of continuous print of the same pattern.

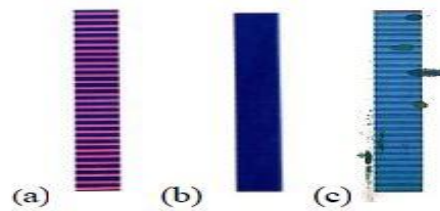


Figure 5. Printed CNT strips on paper from Inkjet printer with water +DMF based ink. (a) 5 runs (b) 10 runs (c) 15 runs.

Figure 5 gives us the output from water and DMF based ink. It can be seen that as the ink is used more i.e. moving from 5 runs to 15 runs, the CNT begin to clog the nozzle. Such agglomerations can be reduced by the effective use of centrifuge.

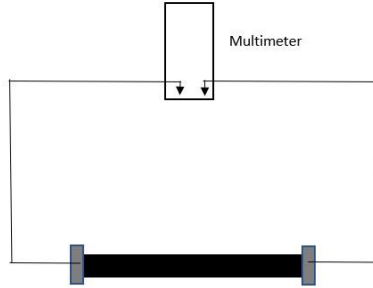


Figure 6. Schematic showing measurement circuit for sheet resistance with silver conductive paint.

A multimeter with two-probes is used to measure the electrical resistance of the printed sheets and calculated the effective sheet resistance. Initially 10 values are taken for each strip by varying the position of the electrodes along the width of the strip to get a good mean value and study the range.

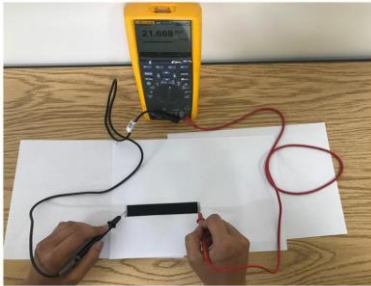


Figure 7. Schematic showing measurement circuit for sheet resistance with silver conductive paint.

A silver conductive paint is applied on each end of the strip along its width to act as an electrode. It is allowed to dry before measurements are taken. The multimeter probe is placed here to check continuity and measure electrical resistance.



Figure 8. Application of silver conductive paint on printed CNT strip.

The mean values for the 10-layers is $4.0133 \text{ M}\Omega$ and the mean value for 5-layers is $43.0769 \text{ M}\Omega$. Conductivity increases as the number of prints increases.

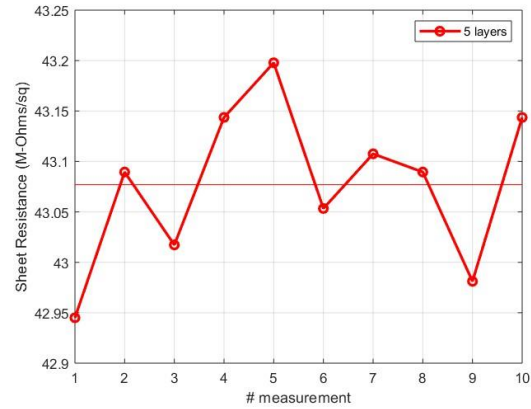


Figure 9. Plot of resistance values measured using multimeter for 5 layers methanol-based CNT.

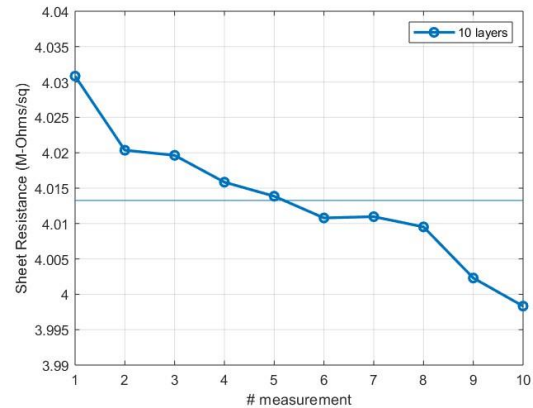


Figure 10. Plot of resistance values measured using multimeter for 10 layers methanol-based CNT.

The results demonstrate that increasing the number of prints to 15 resulted in a tenfold improvement in resistance. It is desirable to achieve a resistance value in the Kilo Ohm range as reported in literature. One explanation for the resistance values being greater than expected is because of the large width of the strip compared to literature. Additionally, this experiment made it clear that 10 layers of print give the best results, a strip with 5 layers gives higher resistance and a strip with 15 layers on a transparency film and highly concentrated CNT methanol-based ink can give a sheet resistance as low as $7.4 \text{ K}\Omega$ [22]. For example, for a strain range of 0.8% we get a gage

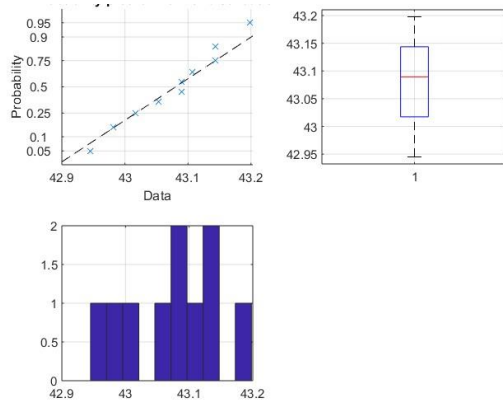


Figure 11. Distribution analysis for 5-layer with Water and DMF based ink

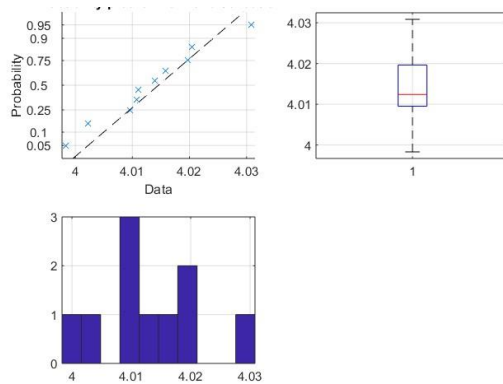


Figure 12. Distribution analysis for 10-layer with Water and DMF based ink

factor of 1.5 [25] under loading for this sheet strain sensor. The resistance values may be further improved by the use of SWCNT since with SWCNT the sheet resistance would be as low as in hundreds of Ohms. Besides this, CNT based strain sensors produced by compression molding can give a gauge factor of as high as 152.93 at 30% strain, whereas commercially available metal-based strain sensors have searchability up to 5% and gauge factor up to 2 [23].

V. CONCLUSION AND FUTURE SCOPE

We reviewed the process of preparing, printing and optimization of carbon nanotubes ink in detail. The research demonstrated inkjet printing as a viable methodology to print CNT conductive networks. For future research, conductive CNT networks and structures may be printed on flexible substrates post surface treatment. Transfer printing may be utilized to imprint the CNT conductive network on integrated circuit packages. Additionally, inkjet-printed CNT

may be used in place of the thermal interface material post inkjet printing and transferring to die prior to integrated heat sink attachment.

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